VERIFAI Specification

AI-Powered Product Verification System — Architecture & Implementation Blueprint

1. Project Overview

Project Name: VERIFAI — AI-Powered Product Verification System

Project Type: Responsive Web Application

Main Objective: To help consumers, inspectors, companies, and authorized administrators assess whether a product appears genuine, suspicious, or cannot be verified using product information, barcode data, packaging images, trusted records, and AI-assisted analysis.

VERIFAI combines AI, OCR, barcode scanning, a trusted product database, deterministic verification rules, and human review with evidence-based reasoning.

2. The Problem VERIFAI Solves

Consumers frequently encounter compromised merchandise, including fake or duplicated packaging, altered expiry dates, invalid identifiers, misleading manufacturer details, mismatched barcodes, or unverified claims. Because shoppers lack direct access to trusted verification tools, VERIFAI bridges this gap securely.

Important limitation: A photograph, barcode, or AI prediction alone cannot prove a product is genuine. VERIFAI communicates uncertainty honestly without making unsupported claims.

3. Proposed Verification Workflow

- **Step 1: Capture or Upload** — Photograph packaging, labels, barcodes, and expiry markers.
- **Step 2: Extract Information** — OCR, barcode scanning, and multimodal AI parse structured observations.
- **Step 3: Compare Trusted Records** — Backend queries authorized product and manufacturer databases.
- **Step 4: Apply Rules** — Deterministic rules evaluate evidence for compliance, warnings, or missing data.
- **Step 5: Display Results** — Provide clear statuses, supporting evidence, limitations, and escalation options.

4. Technology Stack

Frontend: Next.js, TypeScript, Tailwind CSS, shadcn/ui

Backend: Python + FastAPI, Pydantic, SQLAlchemy, Alembic

Database: PostgreSQL relational storage

AI & OCR: Google Gemini API (Flash multimodal), Tesseract OCR, Barcode/QR decoder

Storage: Private S3-compatible object storage

Testing: Pytest and frontend test suites

5. Verification Statuses

- VERIFIED Required identifiers and packaging details match reliable trusted records under configured verification rules.
- SUSPICIOUS Evidence displays a meaningful conflict or documented warning sign requiring review.
- NOT VERIFIED Evidence is insufficient, trusted records are missing, or images are unclear. Never automatically assumed counterfeit.

6. Recommended Implementation Phases

- **Phase 1: Foundation** — Repository inspection, Next.js/FastAPI setup, PostgreSQL, Auth & RBAC.
- **Phase 2 & 3: Data & Scan Workflow** — Product cataloging, secure image upload, OCR, and barcode decoding.
- **Phase 4 & 5: AI & Verification Rules** — Gemini integration with schema validation, deterministic rule execution, and review escalation.
- **Phase 6 & 7: QA, Administration & Deployment** — Dashboards, audit logs, security hardening, and production launch.